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Class: AI

Experiment-4 FAQs

1. What is the difference between a class and an object?

Ans: A class is a blueprint that defines data and functions. An object is an instance of that class with its own data values.

2 Why are real and imaginary data members kept inside the class?

Ans: They are the two parts that together represent a complex number. Keeping them in one class groups related data and operations into a single unit.

3: What does a constructor do?

Ans A constructor is a special member function called automatically when an object is created. It initializes the object's data members.

4. How is the division of two complex numbers calculated?

Ans For $(a+jb)(c+jd)$ multiply by the ~~eq~~ conjugate of the denominator. The result is $(ac+bd)(c^2+d^2)+j(bc-ad)(c^2+d^2)$; division by zero must be checked.

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5. What is a complex conjugate ?

Ans The conjugate of a+jb is a-jb. It keeps the real part unchanged and reverses the sign of the imaginary part.

6 What are data members ?

Ans: Data members are variables declared inside a class to store the state of it's object. In a complex class, they may be real and imag.

7 What are member functions ?

Ans Member functions are functions declared inside a class that define it's behaviour. They can access the class's data members by rules.

8 What is encapsulation ?

Ans Encapsulation combines data and functions that operate inside a class. It helps control how an object's data is accessed and modified.

9. What is data hiding ?

Ans Data hiding restricts direct access to internal data, commonly by declaring data members private. Public member functions provide controlled access.

10 What is abstraction ?

Ans Abstraction shows the essential operations of an object while hiding unnecessary implementation details. For example a user can call add() without need to know it's internal calcution